WEEKLY ASSESSMENT • CLOSED BOOK

Programming & Development

Time Allowed: 45 min
Total Marks: 50
Name: _____

SECTION 1: PROGRAMMING & DATA STRUCTURES

Part A: Pointer and Vector Output Questions (5 Questions)

Q1. What is the output of the following C++ code snippet?

```
int arr[] = {10, 20, 30, 40}; int *p = arr; cout << *(p + 2) << " " << *p++;
```

Output: _____

Q2. Given the following vector operations, write the final contents of vector v.

```
vector<int> v = {1, 2, 3}; v.push_back(4); v.pop_back(); v.insert(v.begin() + 1, 5);
cout << v.size() << " | ";
```

Output & Vector Elements: _____

Q3. Trace the pointer references and determine the values printed.

```
int a = 15, b = 30; int *p1 = &a, *p2 = &b; *p1 = *p2 + 5; p2 = p1; *p2 = 50; cout << a
<< " " << b;
```

Output: _____

Q4. Predict the output of this vector dynamic resizing sequence:

```
vector<int> nums; nums.reserve(5); nums.push_back(10); nums.push_back(20);
nums.resize(4, 99); cout << nums[2] << " " << nums[3] << " " << nums.capacity();
```

Output: _____

Q5. What does the following code display?

```
int val = 8; int &ref = val; ref += 4; int *ptr = &ref; *ptr *= 2; cout << val << " "
<< ref;
```

Output: _____

Part B: Complexity Quiz (5 Questions)

Q6. What is the Time and Space complexity of building a Prefix Sum array of size *N* and performing *Q* range sum queries?

Time Complexity: _____ | Space Complexity: _____

Q7. Analyze the time complexity of the following nested loop structure:

```
for (int i = 1; i <= N; i *= 2) { for (int j = 0; j < i; j++) { // O(1) operation } }
```

Time Complexity: _____

Q8. What is the worst-case time complexity of finding a pair with a given sum in a sorted array using the Two-Pointer technique versus using a nested loop brute-force search?

Two-Pointer Complexity: _____ | Brute-Force Complexity: _____

Q9. State the time complexity of querying the frequency of an element in an unsorted array of size *N* using a Hash Map lookup versus a linear scan.

Hash Map Lookup: _____ | Linear Scan: _____

Q10. What is the auxiliary space complexity of reversing a `std::vector` in-place using two pointers?

Auxiliary Space Complexity: _____

SECTION 2: DEVELOPMENT & CSS

Part A: Flexbox Quiz (6 Questions)

Q11. What is the default value of the `flex-direction` property in CSS Flexbox?

Answer: _____

Q12. Which Flexbox property aligns items along the main axis, and which property aligns items along the cross axis?

Main Axis: _____ | Cross Axis: _____

Q13. Explain the difference between `justify-content: space-between` and `justify-content: space-around`.

Answer: _____

Q14. What does the shorthand property `flex: 1 0 auto;` represent in terms of `flex-grow`, `flex-shrink`, and `flex-basis`?

Answer: _____

Q15. How can you center a child element both horizontally and vertically inside a container using Flexbox? List the required properties.

Answer: _____

Q16. What is the purpose of the `align-self` property, and how does it relate to `align-items`?

Answer: _____

Part B: Practical Layout Challenge (20-Minute Practical Task)

Q17. Navbar Layout Reproduction Challenge:
Write the HTML and CSS markup required to reproduce a standard responsive navigation bar. The navbar must contain:

- A brand logo on the far left.
- A center-aligned set of navigation links (Home, Features, Pricing, About).
- A CTA button ("Sign Up") aligned on the far right.
- All elements vertically centered within a 64px height bar using Flexbox.

SECTION 3: RETENTION CHECK (WEEK 1 REVIEW - DAY 14)

Q18. What is the difference between pass-by-value and pass-by-reference in function parameters?

Answer: _____

Q19. Name the primary CSS box model components from innermost to outermost.

Answer: _____

Q20. What is the time complexity of searching an element in a balanced binary search structure vs an unsorted linear array?

Answer: _____

Q21. How does the `position: absolute` declaration behave relative to its parent containers?

Answer: _____

Q22. Define arithmetic progression sum formula and state how prefix sums help compute range sums in O(1) time.

Answer: _____